

# Day 5 — Directional Evidence, Causal Reasoning & Context- Aware Anomalies

From ‘is it strange?’ to ‘what changed first, given which context?’

# Problem statement

By Day 4 we can decide *this window looks suspicious*. Operations still asks four harder questions:

1. **What changed first?** — temporal precedence among correlated metrics.
2. **Which signal might be upstream?** — directional structure, not just co-movement.
3. **Real anomaly, or expected under the current regime?** — context-aware baselines.
4. **Worth an analyst's time?** — investigation-ready evidence, not raw alarms.

Day 5 is where multivariate anomaly detection meets **causal reasoning** and **context-conditioned scoring**. We don't claim causal proof — we add a layer of formal directional + contextual evidence on top of detection.

# Definition — correlation, directionality, causation

**Correlation** between  $X$  and  $Y$ :

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1]$$

Symmetric, magnitude only, no time direction. **Co-movement, not causation.**

**Directional evidence** — does the past of  $X$  help predict  $Y$  beyond  $Y$ 's own past?

Asymmetric, time-aware, still **predictive**, not interventional.

**Causation** — what happens to  $Y$  if we **set**  $X$  to a value (intervention)? Requires either an experiment or a structural model. **Beyond observational data alone.**

# Pearl's causal ladder

Each rung answers a **strictly stronger** question than the rung below.

- **3. Counterfactual** — “*What would have happened otherwise?*”  
 $P(Y_{X=x} \mid X = x', Y = y')$ 
  - SCM with explicit noise terms; twin-network reasoning
  - Rarely identifiable from observational time series
- **2. Interventional** — “*What if we change X?*”  $P(Y \mid \text{do}(X))$ 
  - do-calculus, RCTs, SCM-based intervention reasoning
  - Requires a causal model or an actual experiment
- **1. Associational** — “*What co-moves?*”  $P(Y \mid X)$ 
  - Pearson, Spearman, classical OLS, conditional distributions
  - Where most anomaly detectors live today

**Day 5 lives between rungs 1 and 2** — Granger, VAR, and PCMCi give *formal* directional evidence on observational data, but they do **not** implement  $\text{do}(X)$ . Time-Series Anomaly Detection | Day 5

# Granger causality — formal definition

**Definition (Granger 1969).** Series  $X$  Granger-causes series  $Y$  if past values of  $X$  improve prediction of  $Y$  beyond what past  $Y$  alone provides.

Two nested regression models, choosing lag order  $p$ :

$$(\text{R, restricted}): \quad Y_t = \alpha + \sum_{\ell=1}^p \beta_{\ell} Y_{t-\ell} + u_t$$

$$(\text{U, unrestricted}): \quad Y_t = \alpha + \sum_{\ell=1}^p \beta_{\ell} Y_{t-\ell} + \sum_{\ell=1}^p \gamma_{\ell} X_{t-\ell} + v_t$$

**Null hypothesis**  $H_0: \gamma_1 = \gamma_2 = \dots = \gamma_p = 0$  — past  $X$  adds nothing.

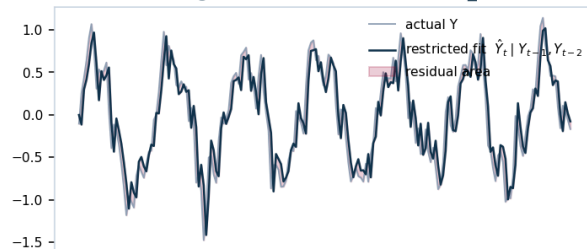
# The F-test in pictures

Granger causality F-test — does adding past X improve the prediction of Y beyond past Y alone?

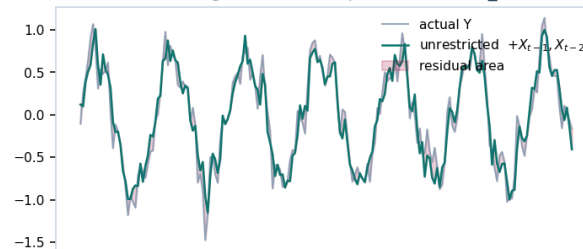
Driver series  $X(t)$  — only its lagged values  $X_{t-1}, X_{t-2}$  enter the unrestricted model



Restricted regression (X excluded) —  $RSS_R = 12.07$



Unrestricted regression (adds past X) —  $RSS_U = 5.70$



$$F = \frac{(RSS_R - RSS_U) / p}{RSS_U / (T - 2p - 1)}$$
$$F = \frac{(12.07 - 5.70) / 2}{5.70 / (198 - 2 \cdot 2 - 1)} = 107.94$$

large  $F \rightarrow$  past X meaningfully improves prediction of Y  $\rightarrow$  reject null  $\rightarrow$  X Granger-causes Y at chosen  $\alpha$

A large  $F$  means the unrestricted model fits substantially better — past  $X$  carries unique predictive information.  
**Reject  $H_0$  at level  $\alpha$ :** there is formal Granger evidence for  $X \rightarrow Y$  at lag order  $p$ .

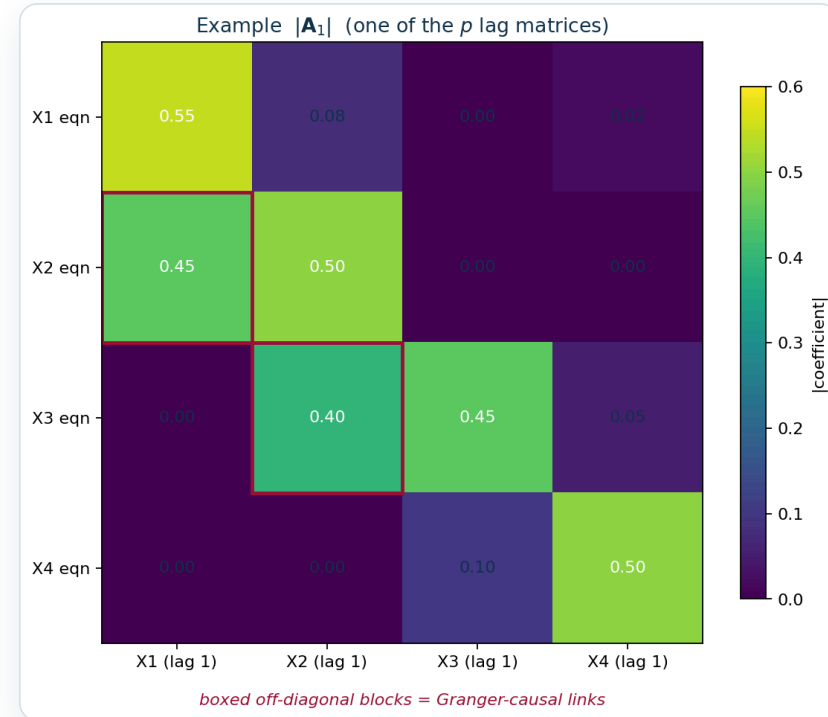
# VAR(p) — the model behind multivariate Granger

Vector AutoRegression of order  $p$ :

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \boldsymbol{\varepsilon}_t$$

with  $\mathbf{y}_t \in \mathbb{R}^d$ ,  $\mathbf{A}_\ell \in \mathbb{R}^{d \times d}$ ,  $\boldsymbol{\varepsilon}_t \sim \square(\mathbf{0}, \boldsymbol{\Sigma})$ .

- $d$  channels,  $p$  lags  $\Rightarrow 1 + dp$  parameters **per equation**,  $d(1 + dp)$  total
- Granger test for  $X_j \rightarrow X_i$  = joint  $F$ -test that  $\{[\mathbf{A}_\ell]_{ij}\}_{\ell=1}^p = 0$
- choose  $p$  by **AIC** or **BIC**; check stationarity
- condition on observed covariates with **Conditional Granger** when feasible



# The synthetic 3-metric panel

$$m_1[t] = 0.80 m_1[t-1] + 0.18 \sin(t/22) + \text{ctx}[t] + \varepsilon_1[t]$$

$$m_2[t] = 0.58 m_2[t-1] + \underbrace{0.34}_{\text{cross}} \underbrace{m_1[t-1]}_{\text{cross}} + 0.10 \text{ctx}[t] + \varepsilon_2[t]$$

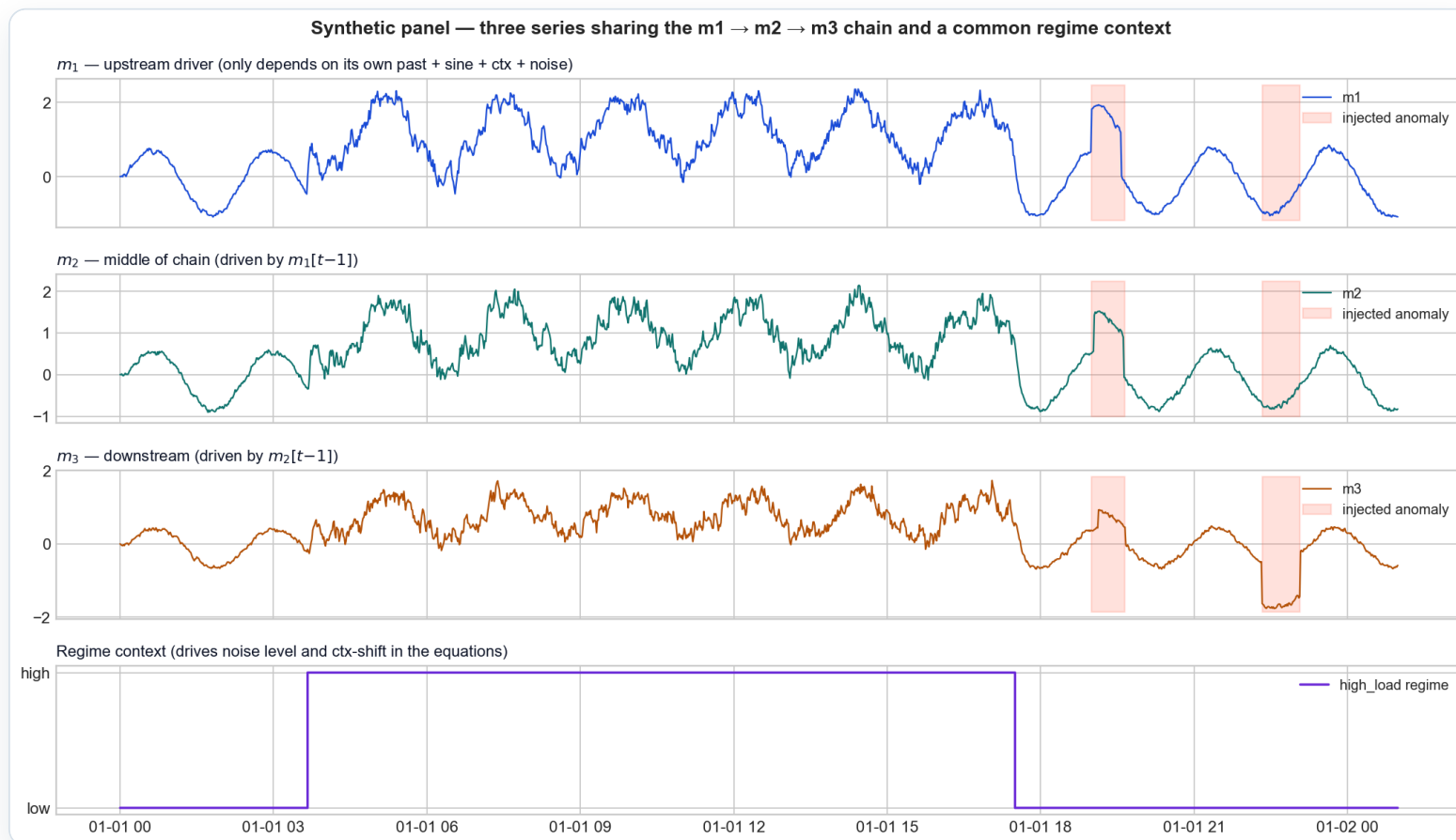
$$m_3[t] = 0.52 m_3[t-1] + \underbrace{0.36}_{\text{cross}} \underbrace{m_2[t-1]}_{\text{cross}} + 0.07 \text{ctx}[t] + \varepsilon_3[t]$$

- **Chain:**  $m_1 \rightarrow m_2 \rightarrow m_3$  (only forward cross-terms)
- **Context**  $\text{ctx}$ : high-load in training, low-load in held-out
- **Anomalies** injected late and propagated with decreasing amplitude

Everything in the next slides — correlation, Granger, VAR residuals, context-aware thresholds — is run against this same panel, so we know what the “right answer” looks like



# What the panel actually looks like



Three series sharing the chain look near-identical to the eye — they are highly *correlated*, but only  $m_1$  leads,  $m_2$  follows, and  $m_3$  follows  $m_2$ . The injected anomalies (red shading, far right) propagate down the chain with decreasing amplitude. The regime panel (purple) shows the high-load training stretch — the noise scale and ctx-shift switch when the regime switches.

# Granger assumptions — and when they fail

| Assumption                | If violated, the matrix can...             |
|---------------------------|--------------------------------------------|
| Linear, additive noise    | miss nonlinear coupling (use PCMCI)        |
| Covariance stationarity   | mix regimes into one misleading fit        |
| Correct lag order $p$     | miss the real precedence horizon           |
| No unmodelled confounders | report spurious links from a common driver |
| Stable sampling cadence   | invent or hide lead/lag artifacts          |

A rejected Granger null is **predictive evidence**, not interventional proof. Treat strong off-diagonal cells as **upstream candidates worth investigating**.

# Preprocessing — what makes the matrix trustworthy

Before running any directional test:

1. **Time alignment** — resample to a common grid, handle clock skew explicitly.
2. **Missing data** — imputation can manufacture spurious dependencies.
3. **De-seasonalize / detrend** — strong periodicities dominate VAR fits unless modelled.
4. **Check stationarity** — ADF / KPSS, or segment by regime before fitting.
5. **Pick  $p$**  — minimize AIC/BIC; sanity-check the value.

The single most common mistake: running Granger / VAR on a panel that spans a deployment, holiday, or autoscaler event. The matrix then describes **two regimes glued together**, not the dynamics of either.

# PCMCI — when Granger / VAR breaks down

**The failure mode.** *Pairwise* Granger ( $X \rightarrow Y$ ) conditions only on the past of  $Y$  — so if a third metric  $Z$  drives both, it reports a **spurious**  $X \rightarrow Y$  link. On our panel:  $m_1$  and  $m_3$  are highly correlated, but  $m_3$  is really driven by  $m_2$ . Pairwise Granger sees  $m_1 \rightarrow m_3$  anyway.

**Conditional Granger** fixes this by conditioning on the past of *every* other channel. But with  $d$  channels and lag  $p$  that means  $d \cdot p$  conditioning variables per test — power collapses when  $d$  is large.

We need the *right* conditioning set per pair — not “nothing” (pairwise) and not “everything” (full conditional).

**PCMCI’s idea** (Runge et al. 2019) — *adapt* the conditioning set to each pair, in two passes:

1. **PC-step (pruning)**. For each target  $Y$ , start with all lagged variables as candidate parents; iteratively drop the ones that become independent of  $Y$  given the parents kept so far. Output: a small estimated parent set  $\widehat{\text{Pa}}(Y)$  per variable.
2. **MCI-step (retest)**. For each candidate edge  $X_{t-\tau} \rightarrow Y_t$ , test

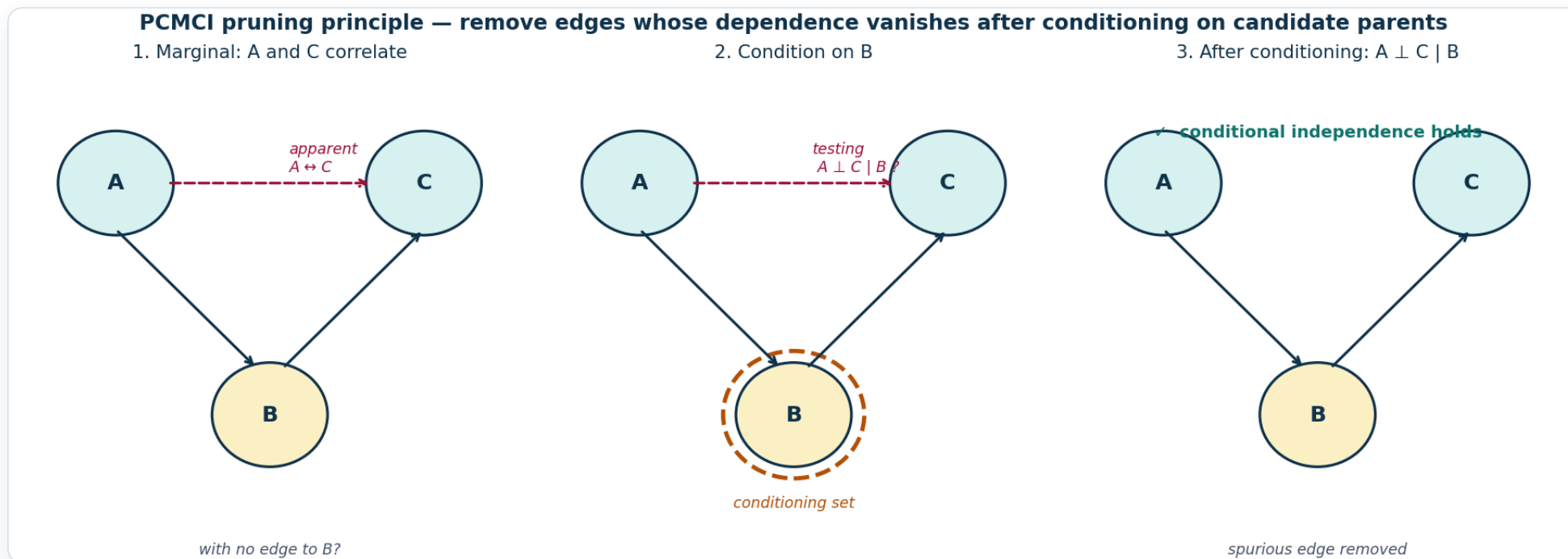
$$X_{t-\tau} \perp\!\!\!\perp Y_t \mid \widehat{\text{Pa}}(Y_t) \setminus \{X_{t-\tau}\}, \widehat{\text{Pa}}(X_{t-\tau}).$$

Conditioning on parents of *both* endpoints kills confounders the PC-step missed.

On the  $m_1 \rightarrow m_2 \rightarrow m_3$  panel, PCMCi's PC-step puts  $m_2$  into  $\widehat{\text{Pa}}(m_3)$ , then the MCI-step finds  $m_1 \perp\!\!\!\perp m_3 \mid m_2$  and **drops the spurious  $m_1 \rightarrow m_3$  edge automatically** — without you having to write that conditioning by hand.

**Extensions: PCMCi+** (contemporaneous links), **LPCMCi** (latent confounders), **JPCMCiplus** (multiple regimes).

# The pruning principle in PCMCI



**Conditional independence test:** if  $A \perp C \mid B$ , the apparent  $A \leftrightarrow C$  link was mediated by  $B$  — remove it. PCMCI generalises this idea across all candidate parents using **ParCorr** (linear-Gaussian) or **CMI**-style nonlinear independence tests.

# PCMCI key hyperparameters

| Parameter            | Meaning                                                    |
|----------------------|------------------------------------------------------------|
| $\tau_{\max}$        | maximum considered lag — runtime + causal horizon          |
| <b>CI test</b>       | ParCorr (linear Gaussian), CMI estimators (nonlinear)      |
| $\alpha_{\text{PC}}$ | significance level inside the PC pruning step              |
| <b>FDR control</b>   | Benjamini–Hochberg over many tests, or family-wise control |

PCMCI is harder to set up than `statsmodels.grangercausalitytests`, but the **right tool when  $d$  is large** or when **nonlinear coupling** is suspected and  $q$  clean conditioning is needed.



# Structural Causal Models (SCMs)

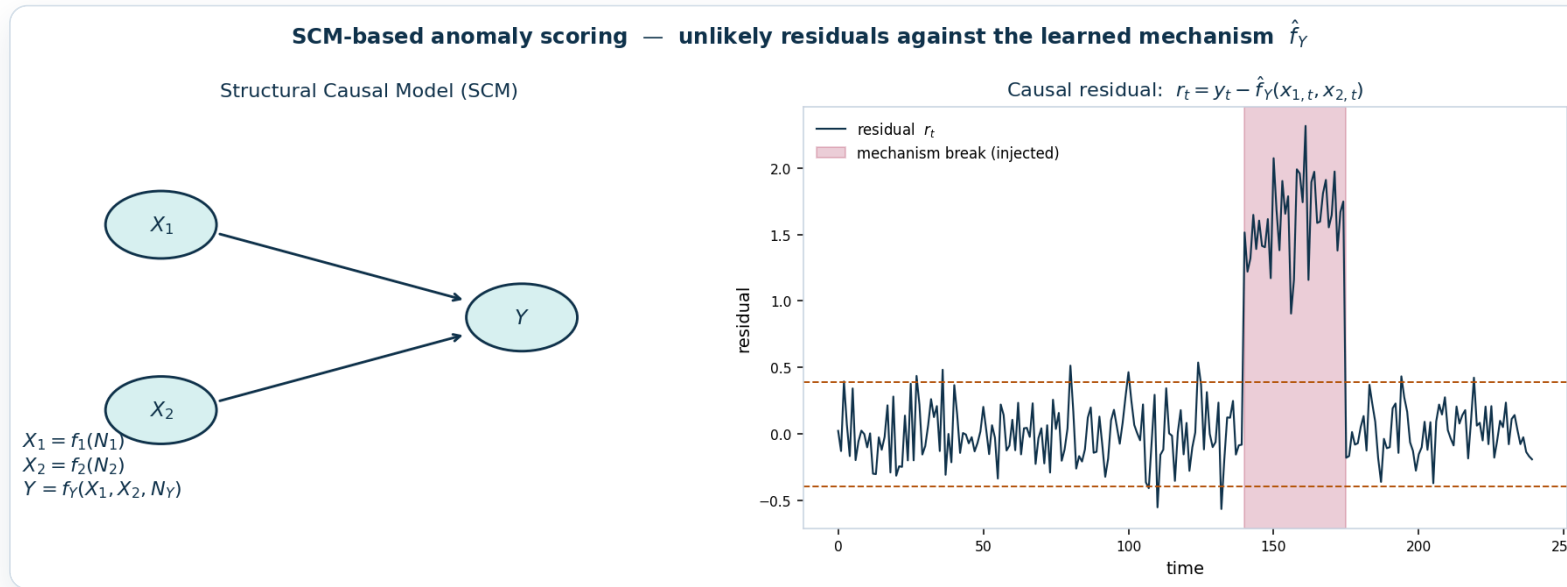
An **SCM** specifies both a directed acyclic graph and explicit generative mechanisms:

$$X_i = f_i(\text{PA}_i, N_i), \quad N_1, \dots, N_d \text{ jointly independent}$$

- $\text{PA}_i$  — direct parents of  $X_i$  in the graph,
- $f_i$  — mechanism (learned or assumed),
- $N_i$  — exogenous noise / unobserved disturbance.

**Why SCMs matter for anomaly detection:** they let us define an anomaly as a **mechanism deviation** — the relationship  $f_i$  failed at time  $t$ , not just the value  $X_i(t)$  was extreme.

# SCM-based residual scoring



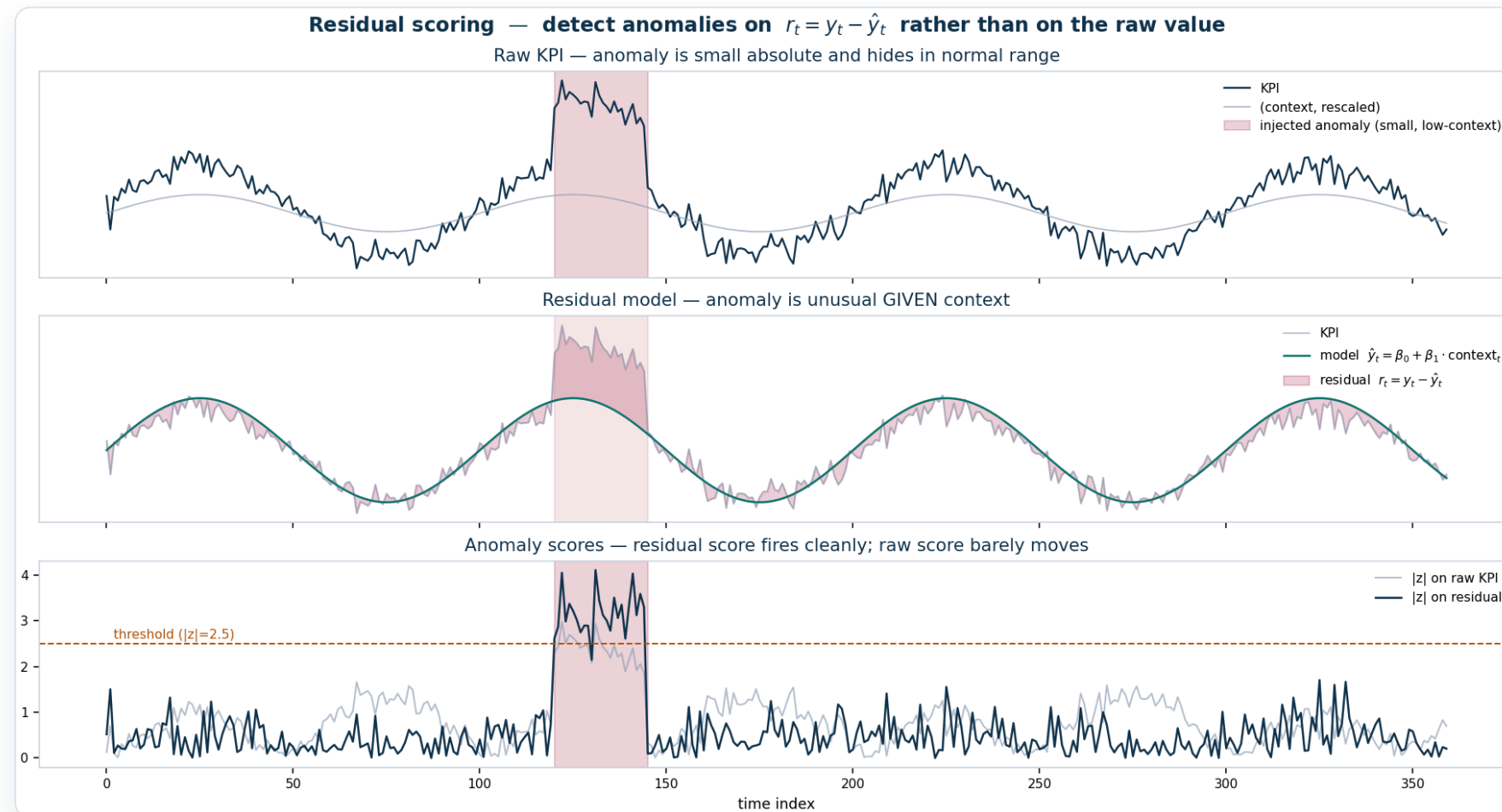
The **causal residual** at time  $t$ :

$$r_t = y_t - \hat{f}_Y(\text{pa}_t)$$

flags  $t$  as anomalous when  $r_t$  is unlikely under the learned noise distribution  $p_{N_Y}^{\hat{}}$ .

**DoWhy / GCM** (Microsoft) operationalises this for industrial use. PCMCi's output graphs can serve as the structure on which SCM mechanisms are fitted.

# Residual vs raw scoring — concept



The same anomaly is **invisible** in the raw KPI score because the absolute deviation is small relative to context-driven variation, but **obvious** in the residual score because the residual model has subtracted the explainable part.

# Two ways to use context

**Feature-level conditioning** — include context variables as predictors:

$$\hat{y}_t = g(Y_{t-1 \dots t-p}, \text{context}_t)$$

Simpler to deploy. Score on  $|y_t - \hat{y}_t|$ . Harder to attribute the anomaly to a single channel.

**Residual modelling** — predict  $Y$  from parents + context, score on residuals:

$$r_t = y_t - \hat{f}_Y(\text{pa}_t, \text{context}_t)$$

More moving parts. Better **explainability** — the residual is a clean “unexplained by the model” signal that supports root-cause discussion.

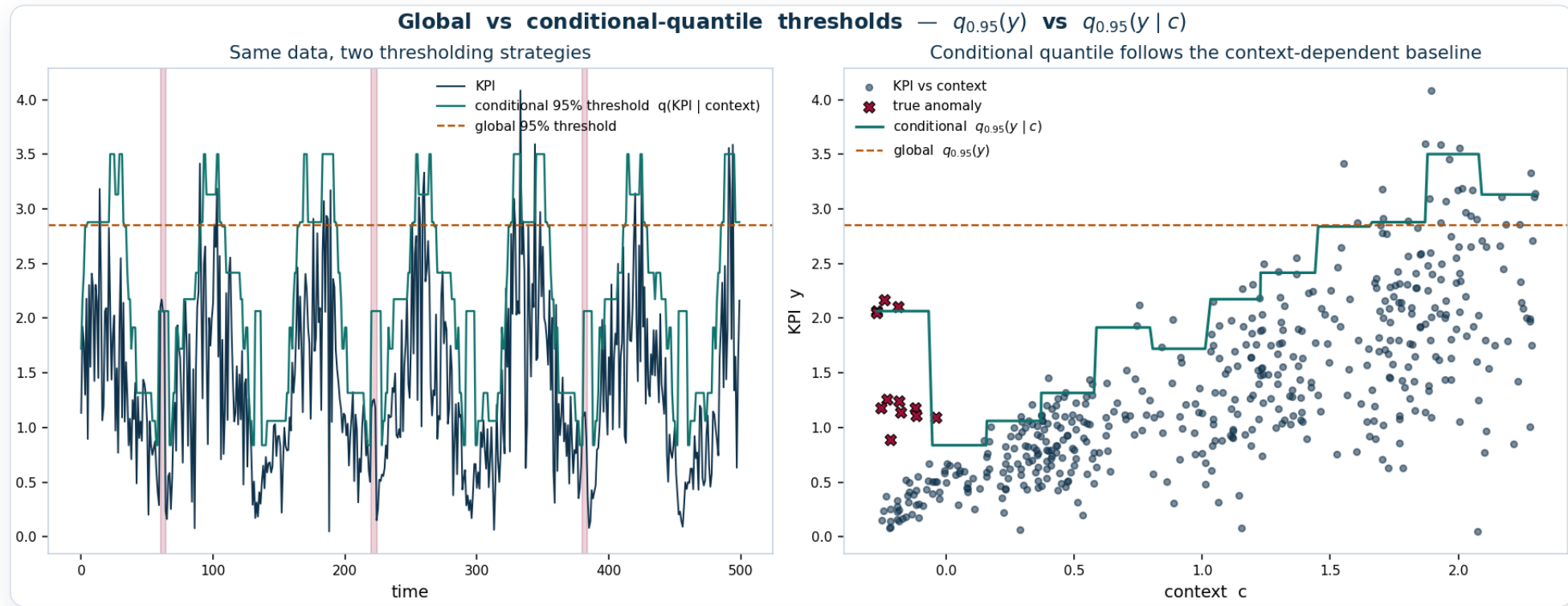
# Context — what we mean operationally

Three recurring kinds of context in operational time series:

- **Load context** — traffic, QPS, concurrent users, queue depth.
- **Shift context** — deployments, config changes, autoscaler actions, holidays.
- **Ambient context** — noisy neighbors, network conditions, diurnal patterns.

A single global threshold ignores all three. A KPI that's high at peak load may be **completely normal**; the same value at 03:00 may be a serious anomaly.

# Global vs conditional-quantile thresholds



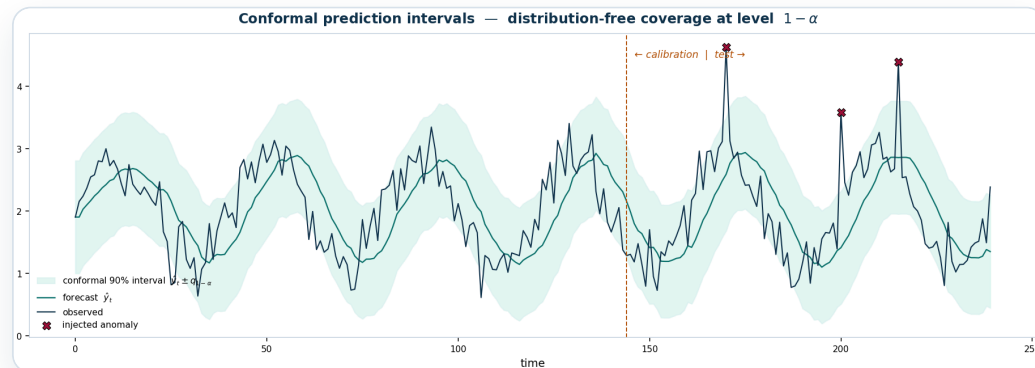
global:  $y_t > q_{0.95}(Y)$  vs. conditional:  $y_t > q_{0.95}(Y | C = c_t)$

The conditional quantile **follows the context-dependent baseline** — it catches anomalies at low load that the global threshold misses, and suppresses the false positives that the global threshold raises during high-load peaks.

# Conformal prediction intervals

A distribution-free alternative to “estimate a quantile”:

1. Split data into **training** and **calibration** sets.
2. Fit any forecaster  $\hat{y}_t$  on training.
3. Compute residuals  $r_i = |y_i - \hat{y}_i|$  on calibration.
4. Take  $q_{1-\alpha}$  of those residuals. The interval  $\hat{y}_t \pm q_{1-\alpha}$  has **marginal coverage**  $\geq 1 - \alpha$ .



Coverage guarantee holds under exchangeability. **Adaptive conformal** (Gibbs & Candès 2021) handles drift by adjusting  $\alpha$  online.

# Regime segmentation + per-segment baselines

When the panel spans multiple regimes:

1. **Change-point detection** (CUSUM, PELT, Bayesian Online CPD) splits the timeline.
2. Fit baseline / thresholds **per segment**.
3. On a new window, decide its segment first, then score against that segment's distribution.

This is the right move whenever **deployments, configuration changes, or seasonal cutovers** create discontinuities you can't simply detrend out. A single global fit will be wrong on both sides.



# From timestamps to events to a triage report

A useful alert review does not dump isolated timestamps. Better pattern:

1. **Detect** windows → score.
2. **Group** adjacent alerts into events.
3. **Inspect** the event peak and dominant residual.
4. **Report** a likely upstream candidate from the directional matrix.

Grouping is not cosmetic — it changes scoring: **event-level / range-aware metrics** match how operations counts hits and misses; per-timestamp F1 misleads when one true event spans many points (see Day 4 on point-adjustment bias).

